

24.11.2016 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
- 2016

Nov-Dec

Max. Marks: 100

Class: SY BTech

Name of the Course: Analysis of Algorithms

Course Code: UITC403

Duration: 3 Hours

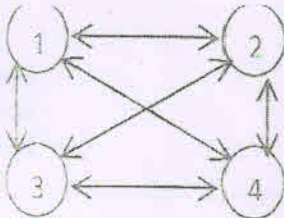
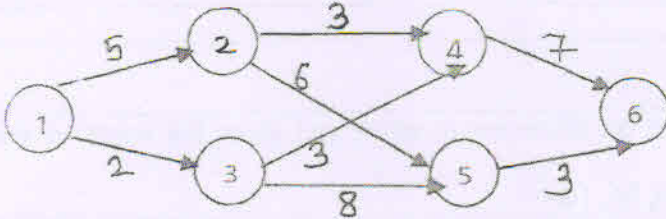
Semester: IV

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Apply Bubble Sort on following numbers and show the result of each pass. <12, 15, 20, 8, 5, 9, 25, 10>	10
Q 1 (b)	Use a recursion tree method to determine a good asymptotic upper bound i.e. Big-Oh (O) value on the recurrence $T(n) = 3T(n/4) + n^2$ and verify it using substitution method. OR	10
Q 1 (b)	Multiply the following integers using Long Integer Multiplication and discuss its complexity. (4578) X (5624)	10
Q2 (a)	Explain Job Sequencing with Deadlines for the given instance. $N=5$, $(p_1, p_2, p_3, p_4, p_5) = (20, 15, 10, 5, 1)$ and $(d_1, d_2, d_3, d_4, d_5) = (2, 2, 1, 3, 3)$	10
Q2 (b)	Write an algorithm for Binary Search and derive its complexity. OR	10
Q2 (b)	Write an algorithm for Merge Sort and derive its complexity.	10
Q3 (a)	Explain Optimal Storage on Tapes problem.	10
Q3 (b)	Find an optimal solution to the following knapsack instance. Number of objects=7, Capacity of the knapsack=15 $(p_1, p_2, p_3, p_4, p_5, p_6, p_7) = (10, 5, 15, 7, 6, 18, 13)$ $(w_1, w_2, w_3, w_4, w_5, w_6, w_7) = (2, 3, 5, 7, 1, 4, 1)$ OR	10
Q3 (b)	Construct the Optimal Binary Search Tree for the identifier set $(a_1, a_2, a_3, a_4) = (\text{char}, \text{float}, \text{int}, \text{while})$ with $(P[1:4]) = (1, 4, 2, 1)$ and $q[0:4] = (4, 2, 4, 1, 1)$	10
Q4 (a)	Write and explain sum of subsets algorithm for $n=6$, $m=30$ and $w[1:6] = \{5, 10, 12, 13, 15, 18\}$	10
Q4 (b)	Write a short note on N-Queens problem. OR	10
Q4 (b)	Write a short note on 15-puzzle problem.	10
Q5 (a)	Explain Longest Common Subsequence with example.	10

Q5 (b)	Find a tour of minimum cost traversed by traveling salesperson for the following graph. <table border="1" data-bbox="763 291 1005 479"><tr><td>0</td><td>10</td><td>15</td><td>20</td></tr><tr><td>5</td><td>0</td><td>9</td><td>10</td></tr><tr><td>6</td><td>13</td><td>0</td><td>12</td></tr><tr><td>8</td><td>8</td><td>9</td><td>0</td></tr></table> OR	0	10	15	20	5	0	9	10	6	13	0	12	8	8	9	0	10
0	10	15	20															
5	0	9	10															
6	13	0	12															
8	8	9	0															
Q5 (b)	Find a minimum cost path from 1 to 6 in multistage graph given below using forward approach. 	10																

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